INCH-POUND

MIL-C-43867B
17 March 1989
SUPERSEDING
MIL-C-43867A
31 December 1985

MILITARY SPECIFICATION

* COUPLERS, CARGO PARACHUTE EXTRACTION SYSTEM, 2, 4, and 7 POINT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers three types of couplers for parachute extraction systems.

* 1.2 Classification. The couplers shall be of the following three types, as specified (see 6.2):

Type _____ 2 point, Link Connector Assembly
Type _____ 4 point, Clevis Cluster Assembly
Type _____ 7 point, Clevis Extraction Assembly

* 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1670

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SPECIFICATIONS

FEDERAL

PPP-B-601 - Boxes, Wood, Cleated-Plywood
PPP-B-636 - Boxes, Shipping, Fiberboard

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

U.S. AIR FORCE

65D3820 - Clevis-Cluster, Assembly Of
67B2209 - Link, Connector-Assembly Of

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

11-1-2941 - Clevis, Extraction Assembly

(Copies of drawings required by contractors are available from the U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014.)

2.2 Non-Government publications. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

* 3.2 Materials, components, and construction. Materials, components, and construction of the type 2-point couplers, Drawing No. 67B2209, and type 4-point couplers, Drawing No. 65D3820, and 7-point couplers, Drawing No. 11-1-2941, shall conform to the requirements specified herein and on the applicable drawings. Materials not definitely specified shall be of the quality normally used by the manufacturer, provided that the completed item complies with all provisions of this document.

* 3.2.1 7-point clevis extraction assembly proofload. The 7-point clevis extraction assembly shall show no visual evidence of fracture, permanent deformation of any component, or defects that interfere with the proper disassembly or assembly of the clevis when subjected to tensile proofloads of 112,500 and 56,250 pounds when tested as specified in 4.4.5.

3.3 Identification marking. Marking for identification shall be as specified on the applicable drawings.

3.4 Workmanship. The load couplers shall be clean and free from sharp edges, burrs, or metal slivers. Components shall not be malformed, damaged, or loose, and shall be correctly assembled. The finish shall be free from evidence of corrosion, breaks through to the base metal, dirt, oil, grease, or other foreign matter.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

* 4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to assure compliance with all dimensional requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.3 and 4.4.4.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

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4.4.2 In-process inspection. Inspection of sub-assemblies shall be made to ascertain that construction details which cannot be examined in the finished product are in accordance with specified requirements. The Government reserves the right to exclude from consideration for acceptance, any material or service for which in-process inspection has indicated nonconformance.

4.4.3 End item visual examination. The end items shall be examined for the defects listed in table I. The lot size shall be expressed in units of couplers. The sample unit shall be one complete coupler. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 4.0 for major defects and 10.0 for total (major and minor) defects.

TABLE I. End item visual defects

Examine	Defect	Classification Major	Classification Minor
Finish	Omitted or not as specified	101	
	Any evidence of corrosion	102	
	Any breaks through coating to base metal	103	
	Supplementary treatment omitted or not as specified		201
	Dirt, oil, grease, or other foreign matter		202
Construction and workmanship	Any component missing	104	
	Any component malformed or damaged	105	
	Sharp edges, burrs, or metal slivers	106	
	Components incorrectly assembled or loose		203
Identification marking	Missing, illegible, incomplete, incorrect, or not permanent		204

4.4.4 End item dimensional examination. The end items shall be examined for conformance to the dimensions specified on the drawings that can be determined on the end items without damaging or disassembling the end items. The lot size shall be expressed in units of couplers. The sample unit shall be one complete coupler. The inspection level shall be S-3 and the AQL for one class, expressed in terms of defects per hundred units, shall be 6.5.

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* 4.4.5 End item testing.

* 4.4.5.1 7-point clevis extraction assembly proofload testing. The 7-point clevis extraction assembly shall be tested as specified in 4.5.1 and 4.5.2. The lot size shall be expressed in units of completed clevis extraction assemblies. The sample unit shall be one completely assembled clevis extraction assembly. The inspection level shall be S-2, and failure of any sample unit shall be cause for rejection of the lot.

4.4.6 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing; damaged, or not as specified.
Workmanship	Inadequate application of components, such as incomplete sealing or closure of flap, improper taping, loose strapping, or inadequate stapling Bulged or distorted container
Content	Number per container is more or less than required

4.4.7 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Interlocking of loads not as specified Load not bonded as specified

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Examine	Defect
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

* 4.5 Methods of inspection.

* 4.5.1 112,500 pound proofload test of the 7-point clevis. The clevis shall be tested in a calibrated tensile testing machine, using a contractor furnished test fixture. A load of 112,500 pounds shall be applied to the single lower attachment point and uniformly distributed among the upper four attachment points as indicated on Drawing No. 11-1-2941. Apply the proofload at a rate from 5,000 pounds/second to 10,000 pounds/second, hold at 112,500 pounds for 15 seconds, and then reduce the load to zero. After proof loading, disassemble the clevis for inspection. Any evidence of fracture, permanent deformation of any component, or defects that interfere with disassembly or assembly shall be cause for rejection of the lot.

* 4.5.2 56,250 pound proofload test of the 7-point clevis. The clevis shall be tested in a calibrated tensile testing machine, using a contractor furnished test fixture. A load of 56,250 pounds shall be applied to the single lower attachment point and uniformly distributed among the upper middle two attachment points as indicated on Drawing No. 11-1-2941. Apply the proofload at a rate from 5,000 pounds/seconds to 10,000 pounds/second, hold at 56,250 pounds for 15 seconds, and then reduce the load to zero. After proof loading, disassemble the clevis for inspection. Any evidence of fracture, permanent deformation of any component, or defects that interfere with disassembly or assembly shall be cause for rejection of the lot.

5. PACKAGING

* 5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A. Each coupler shall be unit packed in a fiberboard box conforming to style RSC (end loading), grade W6c of PPP-B-636. Inside dimensions of each interior box shall approximate the applicable dimensions specified in table II. Each box shall be closed in accordance with method IV as specified in the appendix of PP-B-636.

5.1.2 Commercial. Each coupler shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2).

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5.2.1 Level A packing. Couplers of one type only, preserved as specified in 5.1, shall be packed in a shipping container conforming to overseas type, style A or I, grade A or B, type 2 load of PPP-B-601. The quantity and packing arrangement of the couplers within the container and the approximate inside dimensions of the container shall be in accordance with the applicable requirements specified in table II. Each shipping container shall be closed and reinforced in accordance with the appendix of PPP-B-601.

* TABLE II. Approximate inside dimensions of unit packs and shipping containers, including pack quantity and packing arrangement 1/

Type	Inside Dimensions (inches) Unit Packs	Shipping Containers	Pack Quantity (each)	Packing Arrangements (number)
2-point	5-1/2 x 4-1/4 x 7-1/2	23 x 16 x 11	20	5 x 2 x 2 (on edge)
4-point	8 x 4-1/4 x 9	23 x 16-3/4 x 9-1/2	10	5 x 2 x 1 (upright)
7-point	13-1/2 x 13-1/2 x 9	13-3/4 x 13-3/4 x 9-1/4	1	----

1/ Approximate dimensions are furnished as a guide only.

5.2.2 Level B packing. Couplers of one type only, preserved as specified in 5.1, shall be packed as specified in 5.2.1, except the shipping container shall conform to domestic type, style A or I, grade B, type 2 load of PPP-B-601.

5.2.3 Commercial packing. Couplers, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

* 5.3 Palletization. When specified (see 6.2), couplers of one type only, packed as specified in 5.2.2 or 5.2.3, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet type shall be type I (4-way entry), type IV or type V in accordance with MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding means O or P. Pallet patterns shall be as follows: 2-point, No. 3; 4-point, No. 6; and 7-point, No. 37. Interlocking of loads shall be effected by reversing the pattern of each course.

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5.4 Marking. In addition to any special marking required by the contract or purchase order, shipping containers, unit packs, and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The couplers are used for connecting extraction lines to extraction parachute adapter webs in heavy duty airdrop systems.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this document.
b. Type of coupler required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When first article is required (see 3.1, 4.3, and 6.3).
e. Levels of preservation and packing (see 5.1 and 5.2).
f. When palletization is required (see 5.3).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

* 6.4 Subject term (key word) listing.

Assembly, clevis cluster
Assembly, clevis extraction
Assembly, link connector
System, airdrop

6.5 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-43867B	Couplers, Cargo Parachute Extracting System, 2, 4, and 7 Point

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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